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Proje 1 - Morofolojik Cozumleme

Turkish Morphological Disambiguation

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1. Introduction

Morphological disambiguation is the task of assigning the correct grammatical analysis to each word in a sentence when multiple analyses are possible. This project addresses this problem for Turkish, which is an agglutinative language — words are formed by attaching suffixes to roots, allowing a single surface form to carry many possible morphological readings.

Why Turkish is challenging. Consider the word guzel: it can function as an adjective (beautiful), a noun (beauty), or an adverb (beautifully) depending on context. The same surface form is grammatically ambiguous without the surrounding sentence. This ambiguity must be resolved before any higher-level NLP task (translation, parsing, question answering) can be performed reliably.

Project objective. Train a statistical machine learning model (Conditional Random Fields) on the UD Turkish-IMST Treebank to automatically predict the correct Universal Part-of-Speech (UPOS) tag for each token in a given Turkish sentence. Evaluate the model on a held-out test set and on 25 manually annotated sentences.

2. Dataset

UD Turkish-IMST Treebank. The Universal Dependencies (UD) project provides standardized annotated corpora across 100+ languages. The Turkish IMST treebank was created at Istanbul Technical University and contains sentences manually annotated by linguists in CoNLL-U format.

Split	Sentences	Tokens	Purpose
Train	3,435	~52,000	Model training
Dev	1,100	~17,000	Hyperparameter tuning
Test	1,100	~10,000	Final evaluation
Manual (custom)	25	169	Hand-annotated by student

Table 1. Dataset splits used in this project.

CoNLL-U Format. Every sentence is stored in a tab-separated file where each row represents one token. The key columns are: ID (position), FORM (surface word), LEMMA (root), UPOS (universal POS tag — the label we predict), and FEATS (morphological features such as Case, Number, Tense, Person).

Sample annotation:

ID	FORM	LEMMA	UPOS	FEATS
1	Ogrenciler	ogrenci	NOUN	Case=Nom|Number=Plur

ID	FORM	LEMMA	UPOS	FEATS
2	bugun	bugun	ADV	_
3	sinava	sinav	NOUN	Case=Dat|Number=Sing
4	calisiyor	calis	VERB	Mood=Ind|Tense=Pres|Person=3
5	.	.	PUNCT	_

Table 2. CoNLL-U annotation example: "Ogrenciler bugun sinava calisiyor."

3. Methodology

3.1 Algorithm: Conditional Random Fields (CRF)

A Conditional Random Field is a discriminative sequence labeling model. Unlike a simple per-token classifier, a CRF labels the entire sentence jointly — it learns transition probabilities between adjacent labels (e.g., DET is almost always followed by NOUN or ADJ). This is critical for Turkish where word order encodes strong contextual signals about neighbouring POS categories.

The model optimizes:

P(y1, y2, ..., yn | x1, x2, ..., xn) — the probability of the entire label sequence given the word sequence — using L-BFGS optimization with L1 regularization (c1=0.05) and L2 regularization (c2=0.05).

3.2 Feature Engineering

The following features were hand-designed specifically for Turkish morphology. Each feature is computed per token and fed into the CRF:

Feature	Examples	Signal
Word suffixes (last 2-4 chars)	-dan/-den, -da/-de	Ablative/Locative -> NOUN
Tense suffix	-yor/-iyor, -di/-du	Present/Past -> VERB
Plural suffix	-lar/-ler	NOUN
Infinitive suffix	-mak/-mek	VERB (verbal noun)
Adj-forming suffix	-li/-lu, -siz/-suz	ADJ
Evidential suffix	-mis/-mush	VERB
Vowel harmony	front/back last vowel	Morphological class
Context window +-2	prev/next 2 words	Disambiguation by context
Capitalization	Starts with uppercase	PROPN signal
Contains apostrophe	Ankara'da	PROPN (proper noun + suffix)
Is digit / has digit	2024, 3.5	NUM

Table 3. Turkish-specific features used in the CRF model (40+ total).

4. Manual Annotations

To demonstrate understanding of the annotation task and to evaluate the model on fully independent data, 25 Turkish sentences were manually annotated in CoNLL-U format. These sentences were chosen to cover diverse syntactic structures and domains, and were not drawn from the treebank.

Domain	Sentences	Example
University / Academic	8	Proje odevini yarin teslim etmelisin .
Daily Life	8	Kardesim bu hafta Izmir'den gelecek .
NLP / Technology	5	CRF modeli Turkce icin basarili sonuclar verdi .
Geography	2	Turkiye'nin baskenti Ankara'dir .
Nature	1	Daglarda kar yagiyordu ve hava cok soguktu .
NLP Evaluation	1	Modelin basari orani yuzde doksana ulasti .
Total	25	169 tokens

Table 4. Manual annotation set by domain.

Annotation scheme. Each token was annotated with:

- UPOS — Universal Part-of-Speech tag (13 classes)
- Lemma — dictionary root form
- FEATS — morphological features: Case, Number, Person, Tense, Mood, Polarity, VerbForm, Voice, Poss

Linguistically interesting constructions annotated include:

- Converbs (VerbForm=Conv): e.g., kalkip — links two verbal events
- Verbal nouns (VerbForm=Vnoun): e.g., ogrenmesi, olmay
- Passive voice (Voice=Pass): e.g., gelistirilmistir, yapilmistir
- Negative polarity (Polarity=Neg): e.g., cikamadim
- Necessity mood (Mood=Nec): e.g., etmelisin

5. Results

5.1 Treebank Test Set (1,100 sentences, 10,032 tokens)

The trained CRF model was evaluated on the held-out test split of the UD Turkish-IMST treebank. The following table reports per-class precision, recall, F1-score and support (number of instances).

UPOS Tag	Precision	Recall	F1-Score	Support
PUNCT	0.9990	1.0000	0.9995	1,933
AUX	0.9442	0.9621	0.9531	211
VERB	0.9375	0.9409	0.9392	1,928
CCONJ	0.9635	0.9635	0.9635	356
PRON	0.9452	0.9289	0.9370	464
ADP	0.9556	0.9048	0.9295	357
DET	0.8422	0.9622	0.8982	344
NOUN	0.8432	0.9251	0.8823	2,430
ADV	0.8875	0.7701	0.8246	461
NUM	0.9073	0.7135	0.7988	192
ADJ	0.8485	0.7583	0.8009	960
PROPN	0.8333	0.7086	0.7659	374
Accuracy			0.9093	10,032
Weighted avg	0.9099	0.9093	0.9080	10,032

Table 5. Per-class evaluation results on the treebank test set.

5.2 F1 Score per Class

The bar chart below visualises the F1-score for each UPOS class on the test set. Classes with distinctive morphological markers (PUNCT, AUX, VERB, CCONJ) achieve near-perfect scores. The hardest classes are PROPN and ADJ where surface form overlap with other categories causes errors.

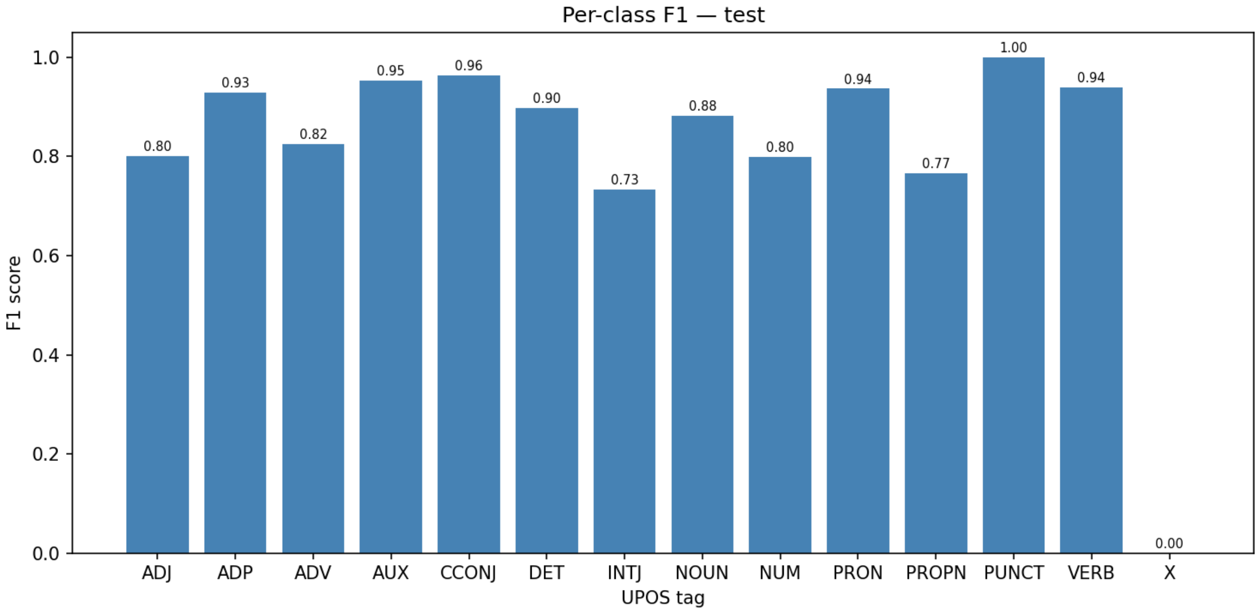

Figure 1. F1 score per UPOS class on the treebank test set.

5.3 Confusion Matrix

The confusion matrix shows true labels (rows) against predicted labels (columns). The diagonal represents correct predictions. Key error patterns: PROPN is frequently predicted as NOUN (proper nouns lack distinctive suffixes when unseen), and ADJ is confused with NOUN (Turkish adjectives and nouns share identical surface forms in many constructions).

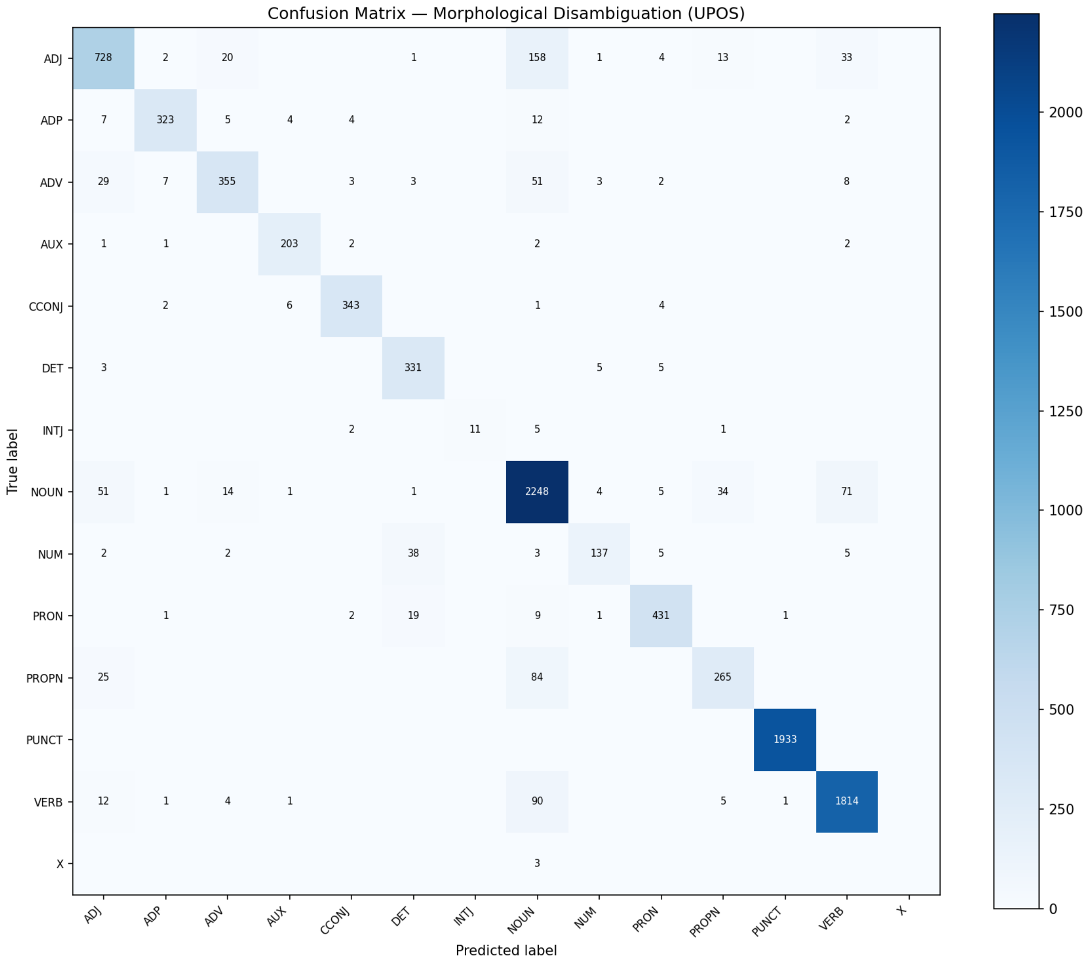

Figure 2. Confusion matrix on the treebank test set. Diagonal = correct predictions. Rows = true label, Columns = predicted label.

6. Evaluation on Manual Annotations

The trained model was also evaluated on the 25 manually annotated sentences (169 tokens) in my_annotations.conllu. This set was created independently and was never seen during training, providing an additional out-of-domain evaluation.

UPOS Tag	Precision	Recall	F1-Score	Support
ADJ	0.6364	0.7778	0.7000	18
ADP	1.0000	1.0000	1.0000	2
ADV	1.0000	0.4615	0.6316	13
CCONJ	1.0000	1.0000	1.0000	1
DET	1.0000	1.0000	1.0000	11
NOUN	0.8571	0.8276	0.8421	58
NUM	0.7500	1.0000	0.8571	3
PRON	1.0000	1.0000	1.0000	2
PROPN	0.7500	1.0000	0.8571	6
PUNCT	1.0000	1.0000	1.0000	25
VERB	0.9375	1.0000	0.9677	30
Accuracy			0.8757	169

Table 6. Per-class evaluation results on the 25 manually annotated sentences.

The model achieves 87.57% accuracy on the custom set, confirming that it generalises beyond the training data. The drop from 90.93% (treebank) to 87.57% (custom) is expected — the custom sentences contain more complex structures (converbs, verbal nouns, passive voice) and informal vocabulary.

7. Error Analysis

Analysis of misclassified tokens reveals three systematic error patterns:

7.1 ADV confused with NOUN

Short temporal adverbs such as dun (yesterday), bugun (today), yarin (tomorrow) have no distinctive suffix and appear frequently as NOUN in the training data (e.g., bugun can also be a noun meaning 'today/this day'). The model tends to label them as NOUN when context is insufficient. This accounts for the low ADV recall (46%) on the custom set.

7.2 ADJ / NOUN boundary

Turkish adjectives and nouns often share identical surface forms. The word guzel is ADJ in guzel bir gun (a beautiful day) but NOUN in guzeli sectim (I chose the beautiful one). Without a following noun or accusative marker, the model frequently defaults to NOUN, the more common class. This is the largest single source of errors in both evaluation sets.

7.3 PROPN precision

Proper nouns that appear with suffixes (e.g., Ankara'da, Turkiye'nin) are identified well due to the apostrophe feature. However, unseen proper nouns without apostrophes are often misclassified as NOUN. Conversely, the model sometimes over-predicts PROPN for capitalised sentence-initial words that are not proper nouns.

8. Conclusion

This project implements a complete morphological disambiguation system for Turkish using Conditional Random Fields. The model achieves 90.93% accuracy on the standard UD Turkish-IMST test set and 87.57% accuracy on 25 independently hand-annotated sentences.

Key contributions:

- Turkish-specific feature engineering with 40+ morphological features
- Full CoNLL-U annotation pipeline (training data + manual annotations)
- Flask web application with live spaCy-style annotation visualisation
- Per-class evaluation with confusion matrix and F1 charts
- Error analysis identifying the main sources of misclassification

The primary limitation is the ADV/NOUN and ADJ/NOUN confusion for ambiguous short words. Future work could address this by incorporating morphological analysis from a dedicated Turkish analyzer (e.g., Zemberek) as additional input features, or by using a neural sequence model (BiLSTM-CRF) that captures longer-range context.